



Tasks

1. Modify the given HTML file to apply **Flexbox** for arranging content inside the page.
2. Apply **CSS Grid** to create a structured layout that includes:
 - Header
 - Navigation
 - Main content
 - Footer
3. Add a **media query** to adjust the layout for smaller screen sizes (mobile view).
4. Insert at least one image and make it **responsive** using:
 - `img` with `loading="lazy"` or
 - the `picture` element

Submission

- Submit your **HTML file** and **CSS file**
- Be ready to demonstrate your page layout in the browser



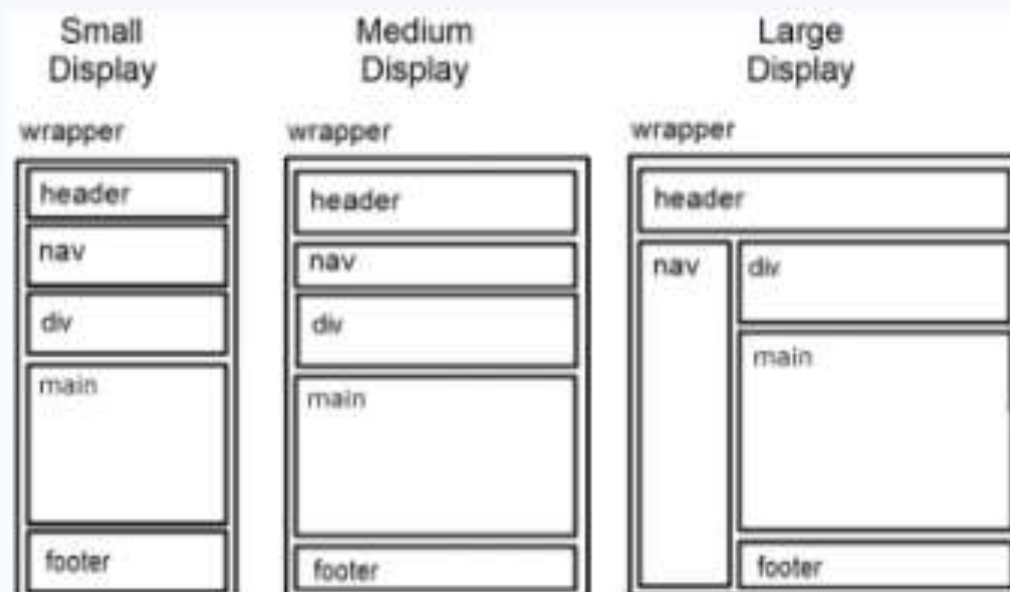
Pacific Trails Resort Case Study

In this chapter's case study, you will use the existing Pacific Trails Resort (**Chapter 7**) website as a starting point to create a new version of the website. The new version will utilize a responsive layout with media queries that displays well on desktop browsers and mobile devices. The updated design features a Mobile First approach, a flexbox navigation layout, and a grid layout for large device display. In addition, you'll modify the content page columns to use flexbox layout instead of float. This is a lot to accomplish but we'll take a step-by-step approach.

First, you will examine the HTML structure of the pages and configure a page layout that works well in smartphones (test with a small browser window). Then you'll resize the browser viewport to be larger until the design "breaks" and code media queries, a feature query, and additional CSS as appropriate using grid layout for the page and flexbox layout for the navigation area. **Figure 8.58** shows wireframes for three different layouts. The Home page web page displays will be similar to **Figure 8.59**.

FIGURE 8.58

Pacific Trails wireframes.





You have five tasks in this case study:

1. Create a new folder for the Pacific Trails Resort website.
2. Review the HTML structure and edit the `pacific.css` external style sheet to configure a single-column (smartphone) display.
3. Configure the CSS needed for pleasing display of the web pages on medium sized mobile devices.
4. Configure the CSS and HTML needed for a pleasing display of the web pages on large mobile devices and desktops.
5. Add a viewport meta tag to each web page.

Task 1: Create a folder called `ch8pacific` to contain your Pacific Trails Resort website files. Copy the files from the **Chapter 7 Case Study** `ch7pacific` folder into the `ch8pacific` folder.

Task 2: Configure a Small Single-Column Layout. Open the `index.html` file in a text editor. View the HTML and notice that a `div` assigned to the `wrapper` id has child elements of `header`, `nav`, `div`, `main`, and `footer` as shown.

```
<div id="wrapper">
  <header> ... </header>
  <nav> ... </nav>
  <div> ... </div>
  <main> ... </main>
  <footer> ... </footer>
</div>
```



a. Configure the CSS. Launch a text editor and open the `pacific.css` style sheet. Edit the styles to achieve a layout that displays well on small devices using normal flow (no floats) with full-width block elements.

1. Remove any style declarations that configure float and left margin from all selectors.
2. Edit the styles for the `nav` element selector. Remove the width declaration. Set padding to 0. Configure centered text.
3. Edit the styles for the `#wrapper` id selector. Remove the width, min-width, max-width, margin, border, and box-shadow declarations.
4. Code a style rule for the `nav li` selector. Set a 1 pixel dark blue solid bottom border.
5. Edit the styles for the header element selector. Remove the style declarations for padding and height.
6. Edit the styles for the `h1` element selector. Remove the declaration for `font-size`. Set top and bottom padding to 0.5em.
7. Remove the section element selector and all style declarations.
8. Save your `pacific.css` file. Use the CSS validator (<https://jigsaw.w3.org/css-validator>) to check your syntax. Correct and retest if necessary.

b. Test the web pages. Display your `index.html` file in a browser. If your browser viewport is a typical size, the page will look a bit awkward and similar to Figure 8.60. This layout is intended for narrow mobile screens. Resize your browser to be narrower until your display is similar to the Small Display shown in Figure 8.59, which simulates mobile display. Test the `yurts.html` and `activities.html` files in a similar manner—they should be similar to the Small Display in Figures 8.61 and 8.62.

**FIGURE 8.60**

Normal flow full-width block elements.

**FIGURE 8.61**

The Yurts page.

**FIGURE 8.62**

The Activities page.





Task 3: Configure a Medium Layout. Display your `index.html` file in a browser—first narrow it and then gradually widen it. The point where it starts to seem awkward in the navigation area is around 600px wide, so that’s what you’ll code for our media query. When the media query is triggered, you’ll configure the layout to follow the Medium Display wireframe in **Figure 8.58**, which has a horizontal navigation bar.

a. Configure the CSS. Launch a text editor and open the `pacific.css` style sheet. Place your cursor below the existing styles. Configure a media query that is triggered when the minimum width is 600px or greater. Code the following styles within the media query.

1. Configure styles for the `nav ul` selector. Configure a flex container for a row that does not wrap. Also configure styles to cause the browser to display empty space before, between, and after the flex items.
2. Configure styles for the `nav li` selector. Eliminate the bottom border (hint: use `border-bottom: none;`).
3. Code a style rule for the section element selector. Set 2em left and right margin.
4. Save your `pacific.css` file. Check your syntax with the CSS validator (<https://jigsaw.w3.org/cssvalidator>). Correct and retest if necessary.

b. Test the web pages. Display your `index.html` file in a browser. You should be able to resize your browser viewport and obtain a display similar to the Medium Display in **Figure 8.59**. When you test the Yurts and Activities pages, you will notice that their Medium Display is still different from **Figures 8.61** and **8.62**.

c. You need to make a few changes to configure the Yurts and Activities pages so that their content within the section elements uses three columns similar to the layout in **Figures 8.61** and **8.62**. A quick way to accomplish this would be to revert back to the **Chapter 7 Case Study CSS**—configure the section element selector with right float and 33% width. Another way to create the three-column look is to configure a flex container for the three section elements, which is what you will do in this case study.

1. Modify the HTML. The `flow` class will be used to configure a flex container. Edit the `yurts.html` file. Code a div assigned to a class named `flow` that contains all section elements. Save the file. Edit the `activities.html` file. Code a div assigned to a class named `flow` that contains all section elements. Save the file.



2. Modify the CSS. Edit the `pacific.css` file. Code a new style rule within the media query for the `flow` class selector.

Configure a flex container. The flex direction is `row`. Add a declaration to the styles for the section element selector and set the `flex` property to 1. Save the `pacific.css` file. Check your syntax with the CSS validator

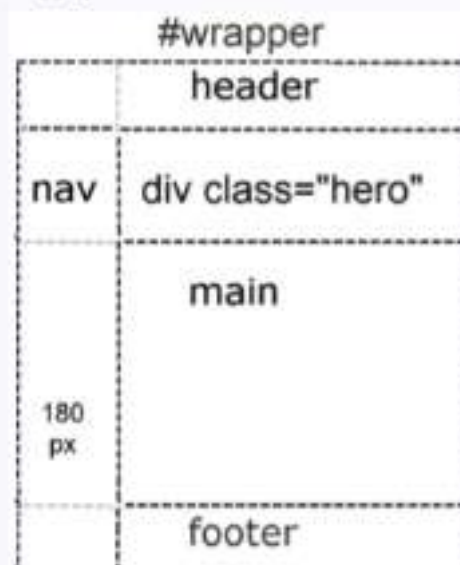
(<https://jigsaw.w3.org/cssvalidator>). Correct and retest if necessary.

3. Display your `index.html`, `yurts.html`, and `activities.html` files in a browser. The display of the `index.html` file should remain unchanged. However, the Yurts page should be similar to Figure 8.61, and the Activities page should be similar to Figure 8.62.

Task 4: Configure a Large Grid Layout. Display your `index.html` file in a browser and gradually widen—you may notice that the page seems to get a bit awkward around 1024px, so that's what you'll code for the next media query. When the media query is triggered, you'll configure the grid layout for supporting browsers in Figure 8.63, which corresponds to the Large Display wireframe in Figure 8.58. Nonsupporting browsers (such as Internet Explorer) will render a web page layout with horizontal navigation.

FIGURE 8.63

Grid layout for Large Display.





a. Configure the CSS. Launch a text editor and open the `pacific.css` style sheet. Place your cursor below the existing styles. Configure a media query that is triggered when the minimum width is 1024px or greater. Code the following styles within the media query.

1. Configure styles for the `#wrapper` selector. Set auto margin, 80% width, a 1px solid dark blue border, and a 3px dark blue drop shadow.
2. Configure styles for the `nav` element selector. Configure left text alignment and 1em left padding.
3. Next, get ready for some progressive enhancement with grid layout. Configure a feature query (use `@supports`) to check for grid display). Configure the following styles within the feature query.
 - i. Configure styles for the `nav ul` selector. Set the `flex-direction` property to the value `column`. Set the top padding to 1em.
 - ii. Notice the grid areas in **Figure 8.63**. Code style rules to configure each child element of the `#wrapper` with the `grid-area` property to correspond to **Figure 8.63**. Use **Hands-On Practice 8.5** as a guide to configuring the header, nav, main, and footer element selectors. Configure the hero class selector as a grid-area as shown:
`.hero { grid-area: hero; }`
 - iii. Code styles for the `#wrapper` selector. Set the `display` property to `grid` and use **Figure 8.63** as a guide to configure the `grid-template` property.
4. Save your `pacific.css` file.

b. Modify the HTML. Launch a text editor and edit the `index.html`, `yurts.html`, and `activities.html` files. The class named `hero` is part of the grid layout. Locate the div that contains the hero image on each page. Add `class="hero"` to the div tag. Save your files.

c. Display your `index.html` file in a browser. You should be able to resize your browser viewport and obtain a layout similar to the Large Display in **Figure 8.58**. The display of the Home page should be similar to **Figure 8.59**. Test the `yurts.html` and `activities.html` files in a similar manner. The Yurts page should be similar to **Figure 8.61**. The Activities page should be similar to **Figure 8.62**.



Task 5: Add a Viewport Meta Tag. Launch a text editor and edit the `index.html`, `yurts.html`, and `activities.html` files. Configure a viewport meta tag in the head section of each page that configures the width to the device-width and sets the initial-scale to 1.0. Save your files. When you test them in a browser, the display will be unchanged, but the viewport meta tag will improve the display on a mobile device.

You have accomplished a great deal as you completed this case study. The design is now responsive and displays well on devices with various size viewports. You have also configured a modern grid layout which includes a flexbox layout navigation area and a flexbox layout with three columns on the content pages. Pacific Trails Resort is responsive and mobile!